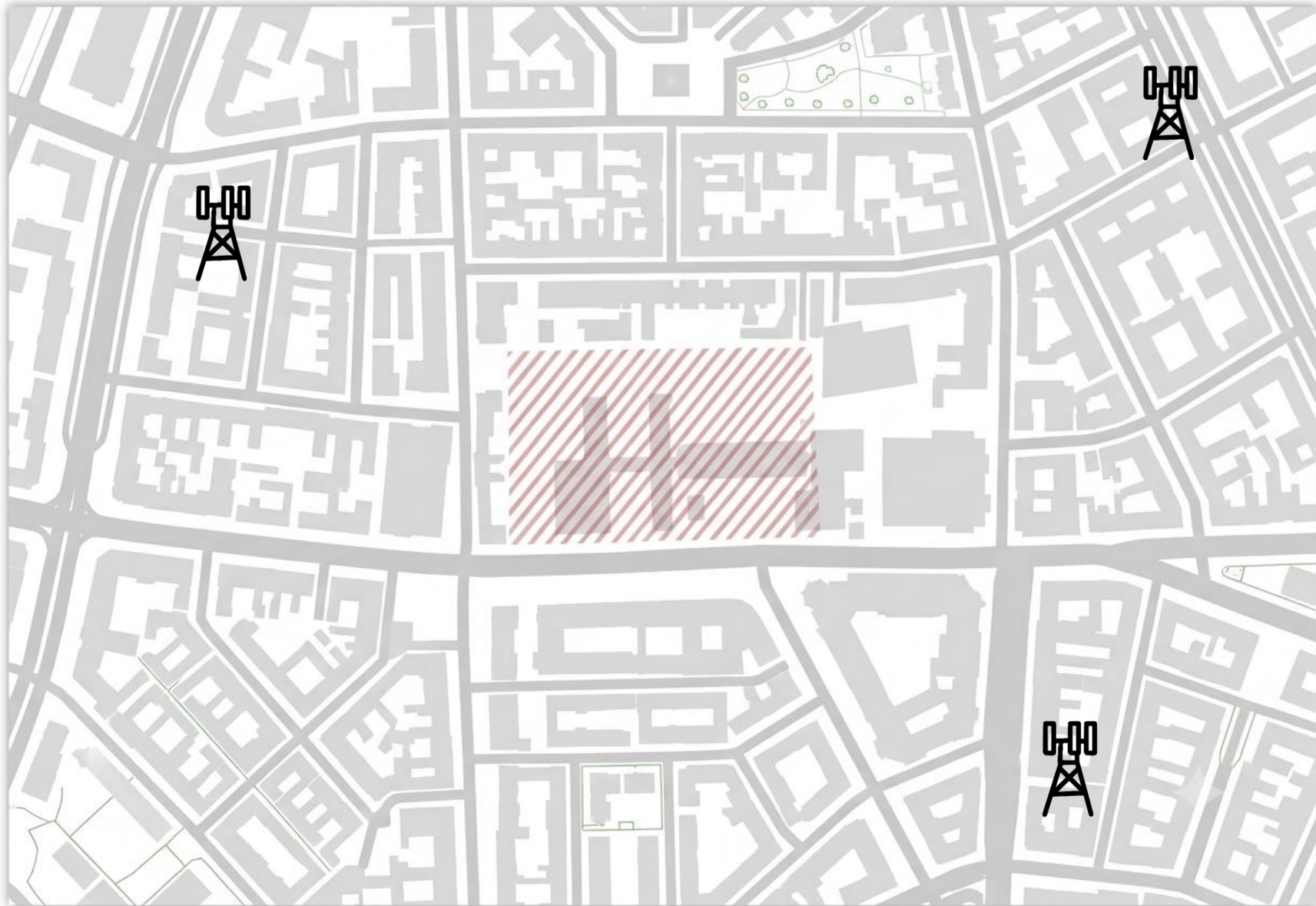
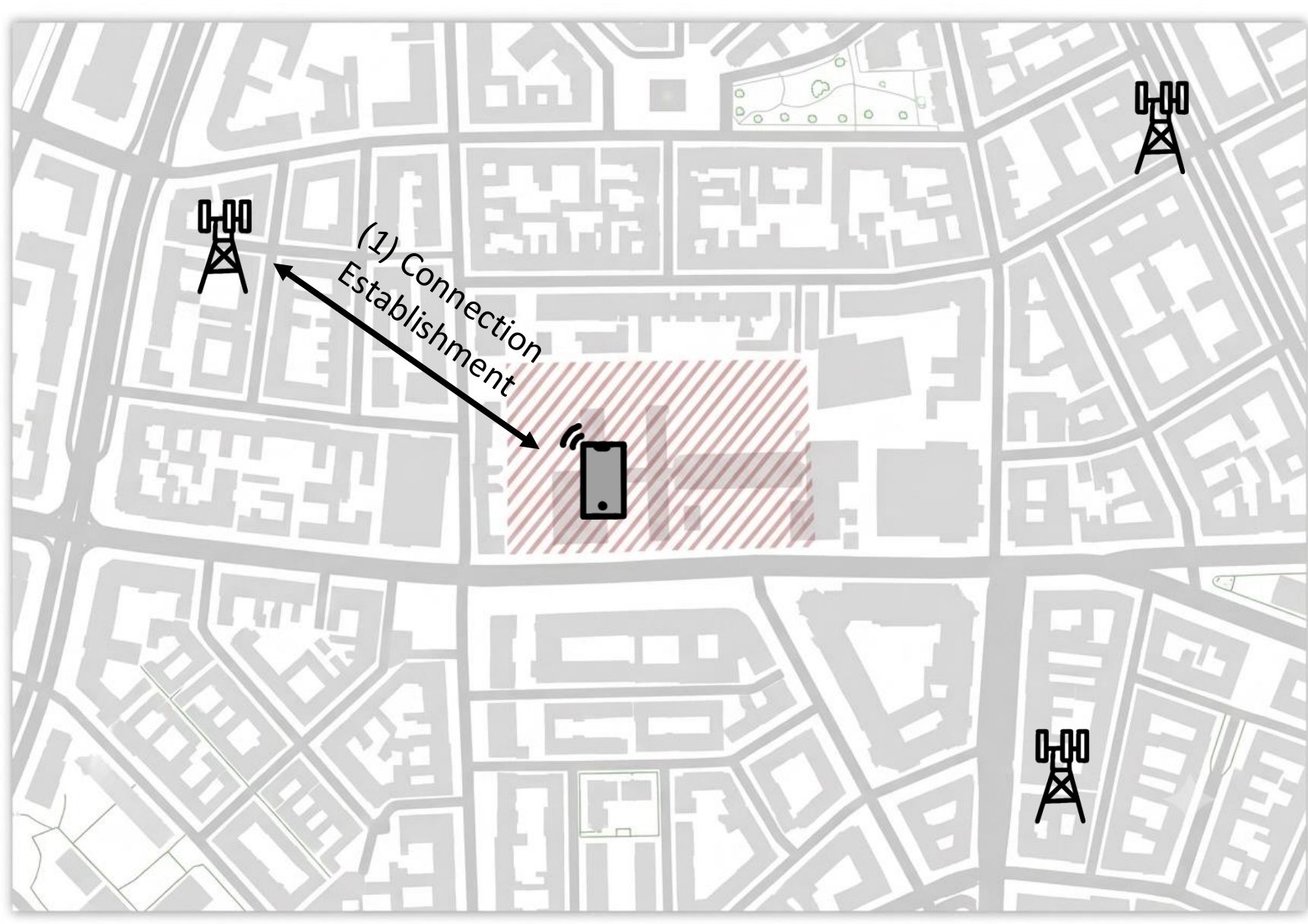


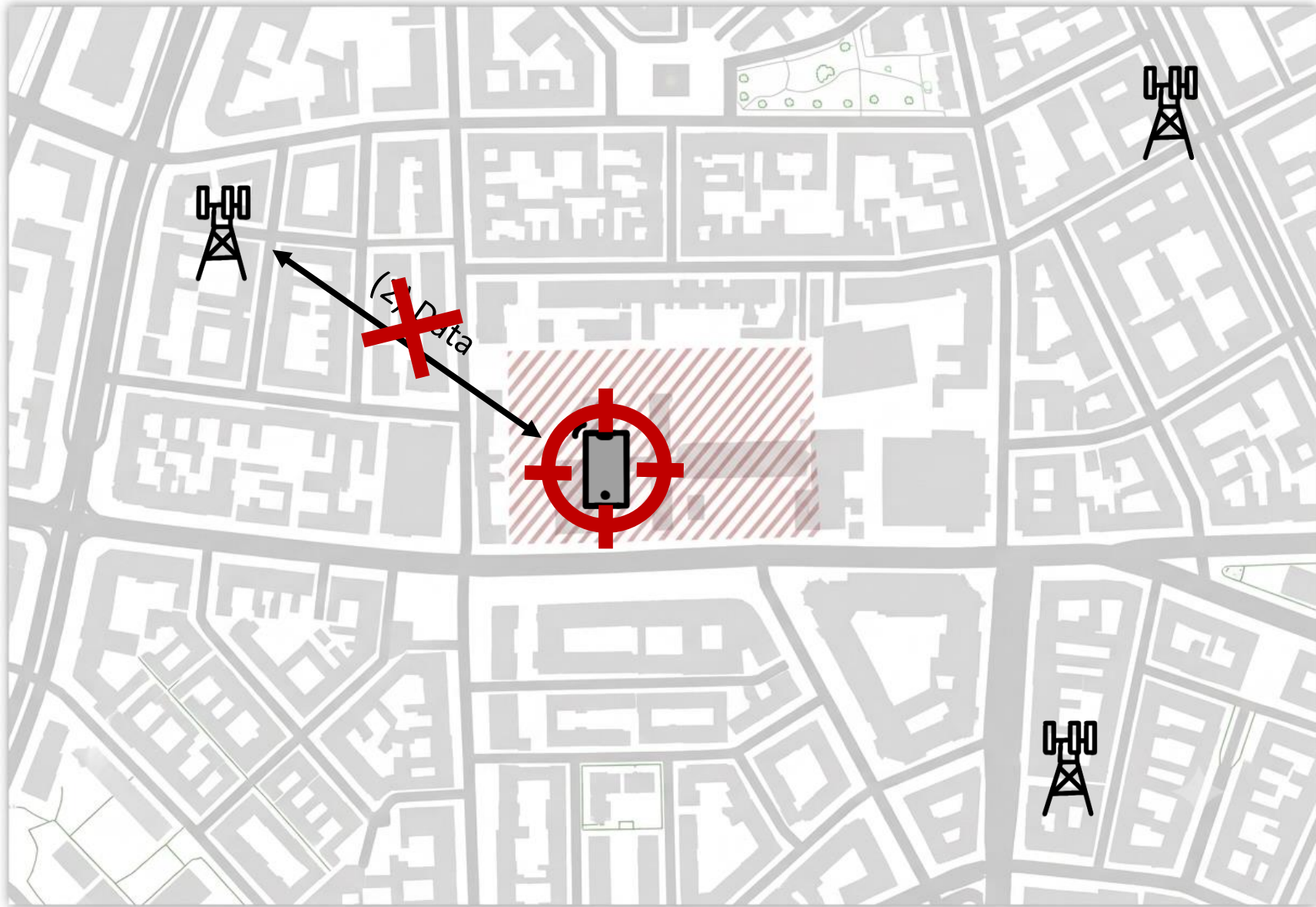
Finding Phones Fast: Low-Latency and Scalable Monitoring of Cellular Communications in Sensitive Areas

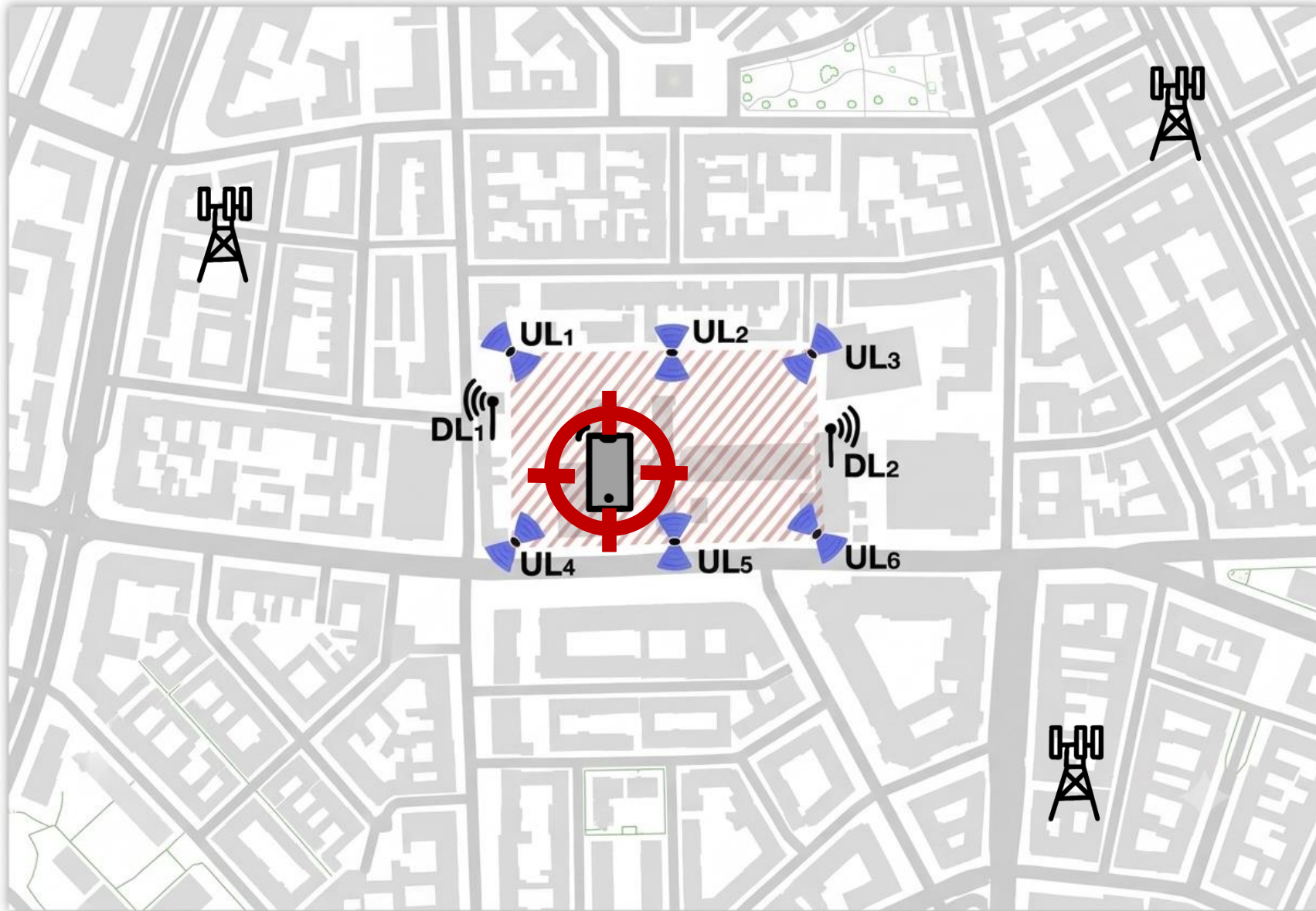
Martin Kotuliak, Simon Erni, Jakub Polak, Marc Roeschlin, Richard Baker, Ivan Martinovic, Srdjan Čapkun



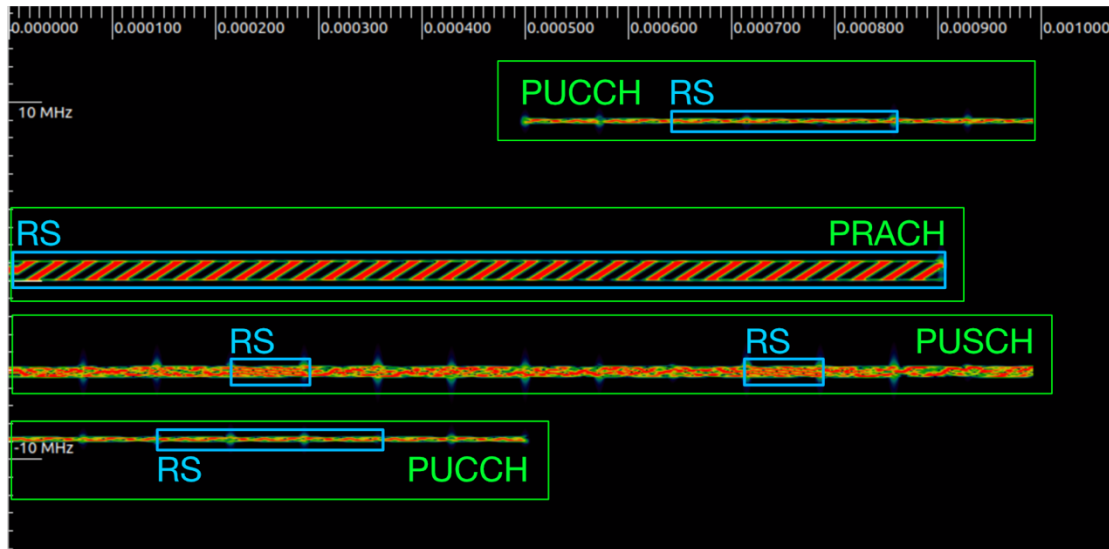








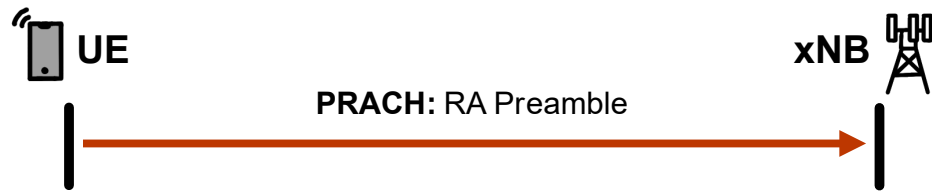
Detection Based on Measurement of Reference Signals



3 uplink messages in Connection Establishment

- 1x PRACH
- 1x PUSCH
- 1x PUCCH

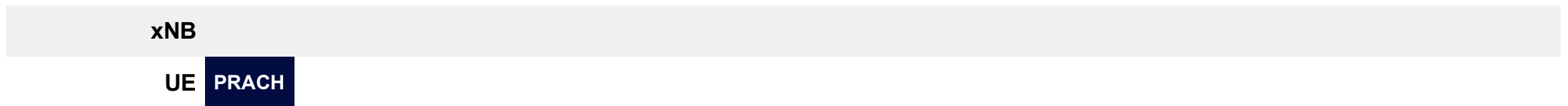
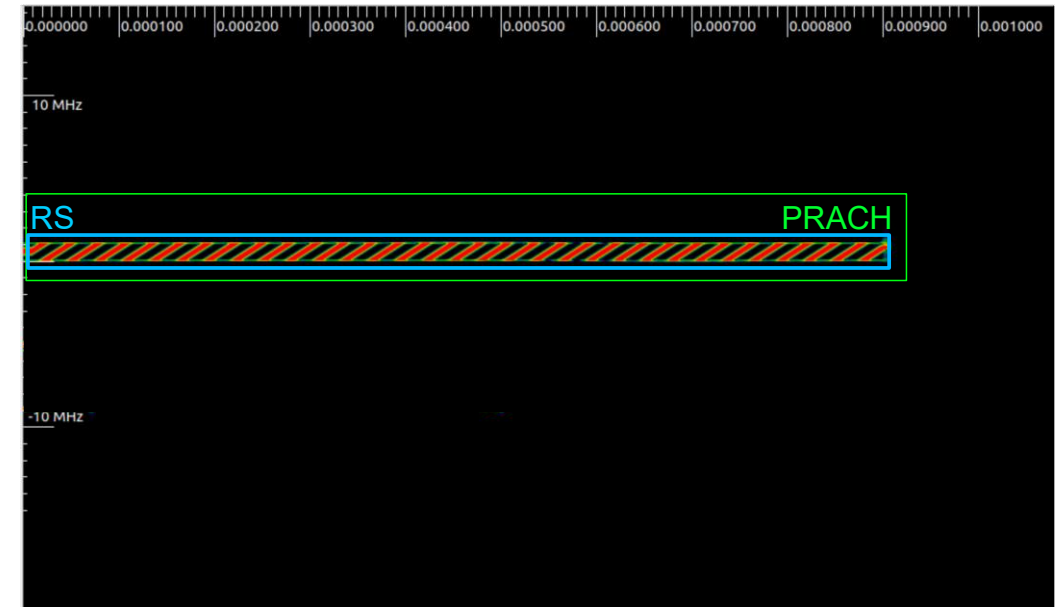
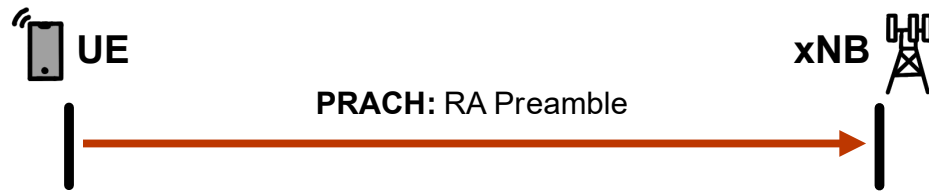
Connection Establishment



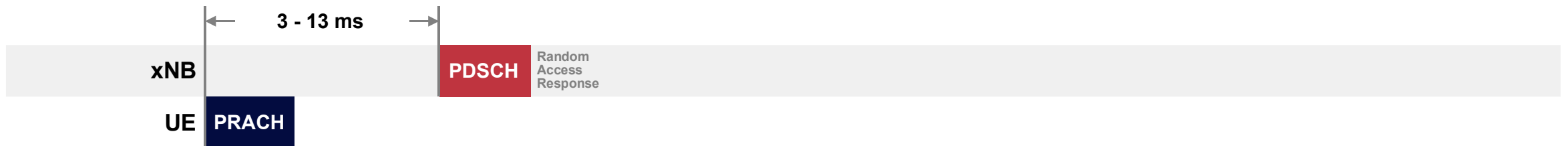
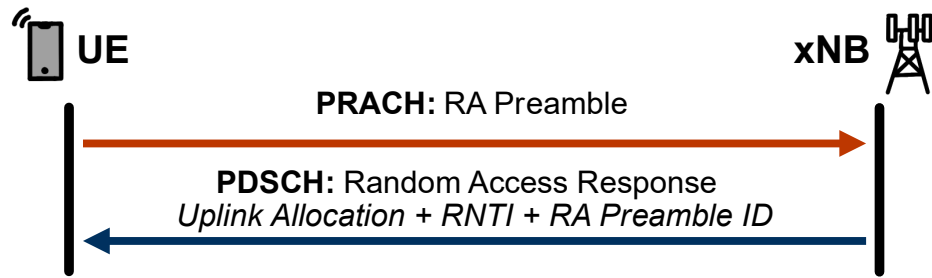
xNB

UE PRACH

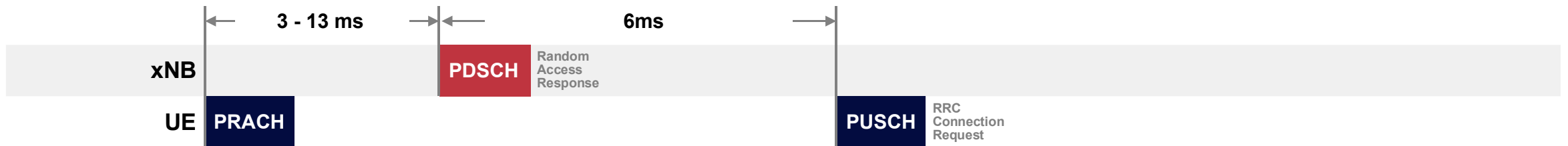
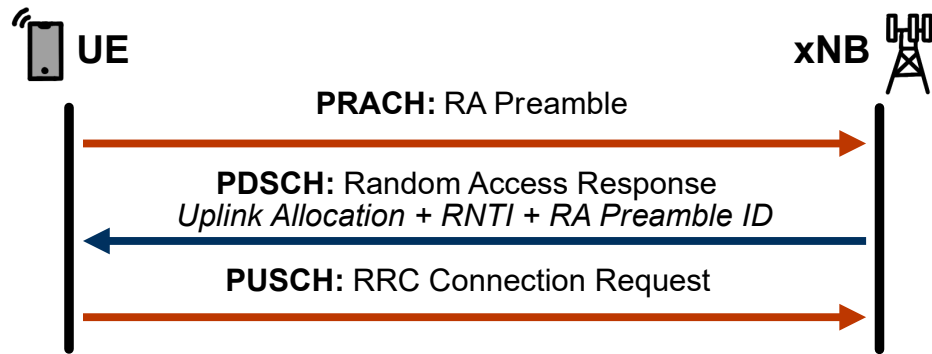
Connection Establishment: PRACH



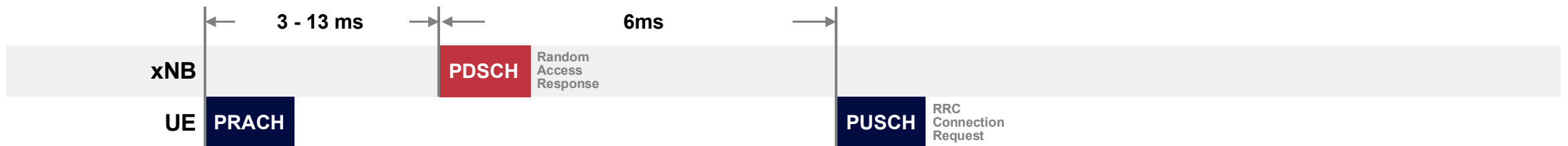
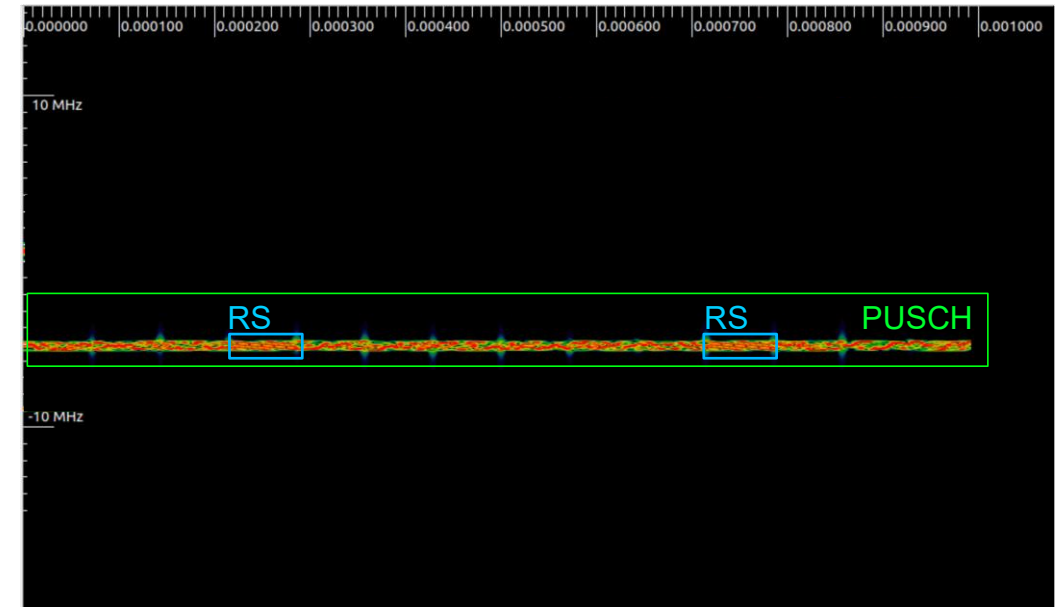
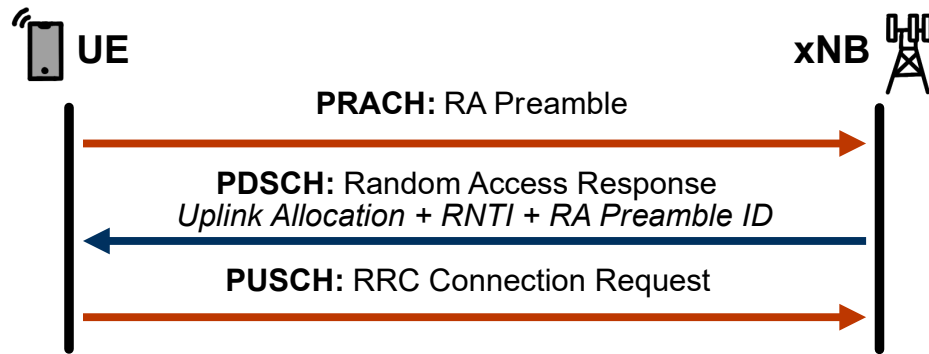
Connection Establishment



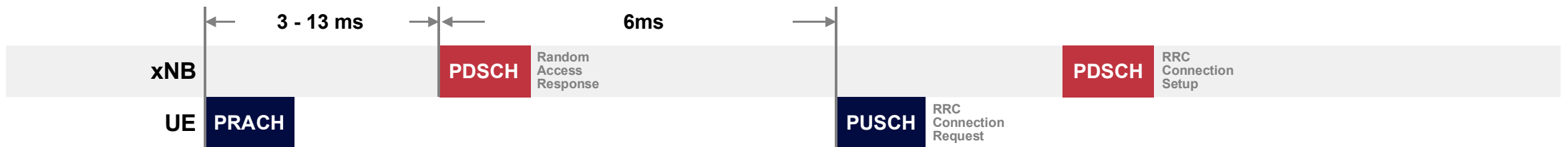
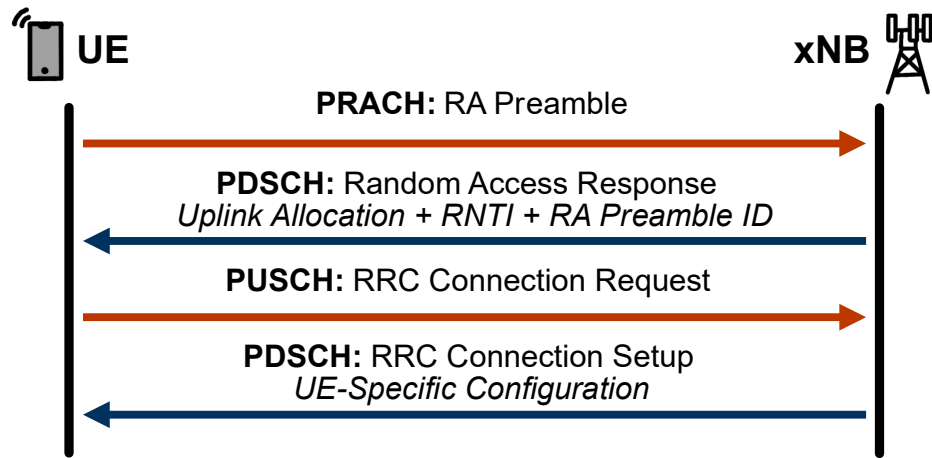
Connection Establishment



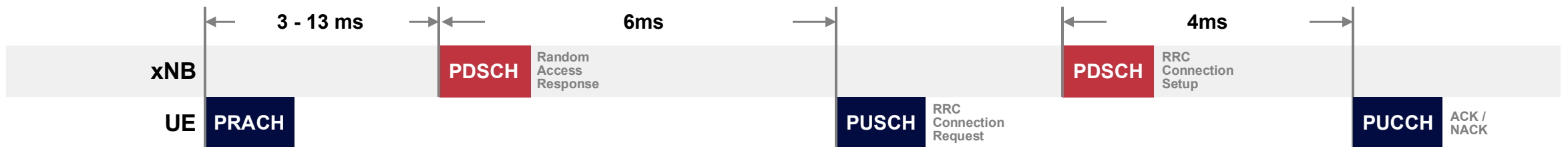
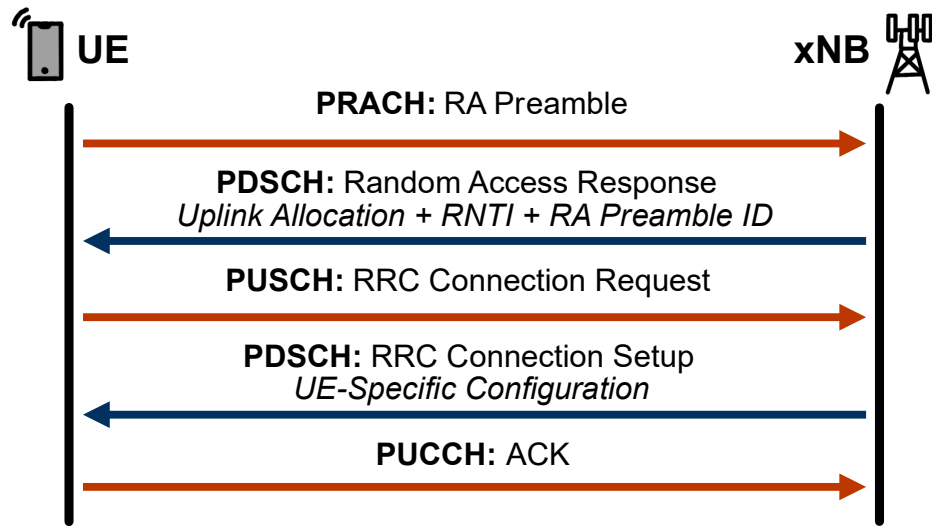
Connection Establishment: PUSCH



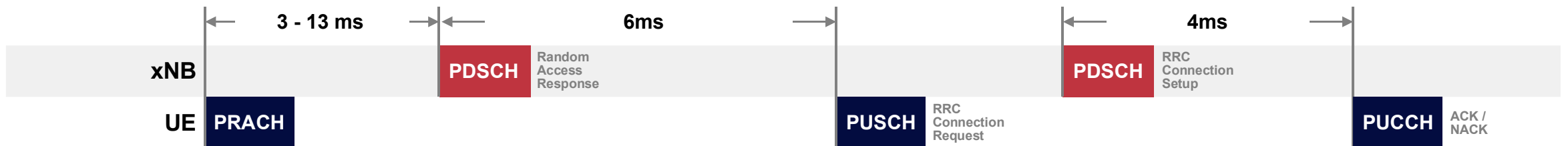
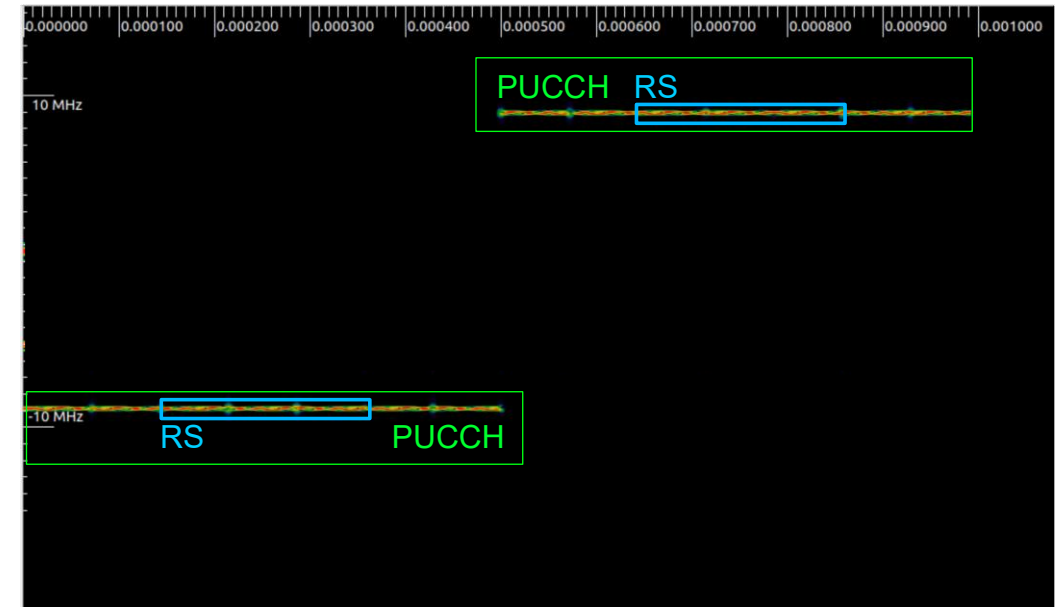
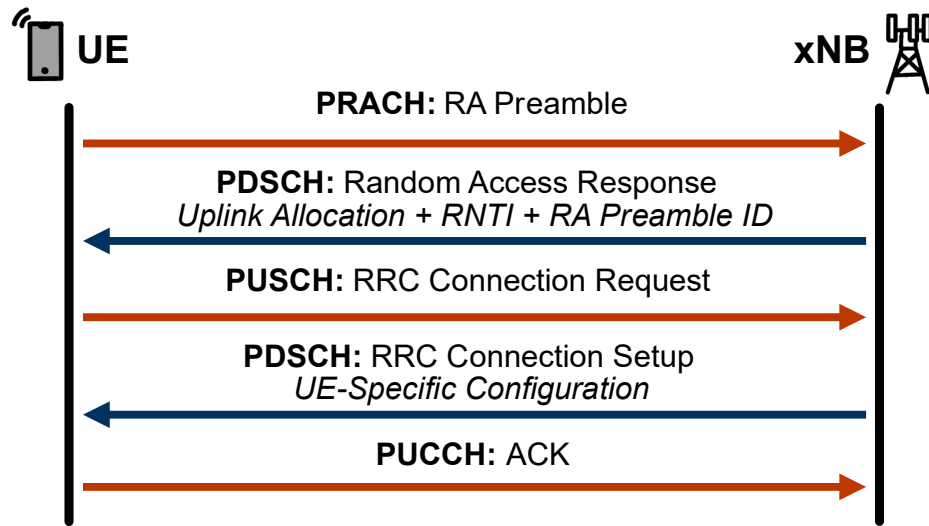
Connection Establishment



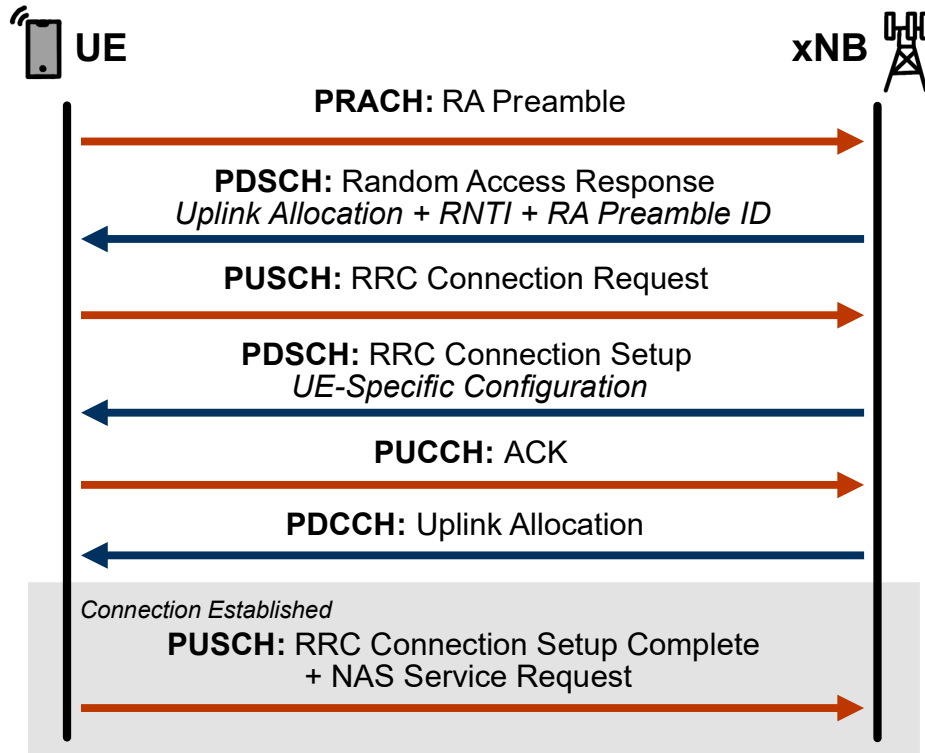
Connection Establishment



Connection Establishment: PUCCH



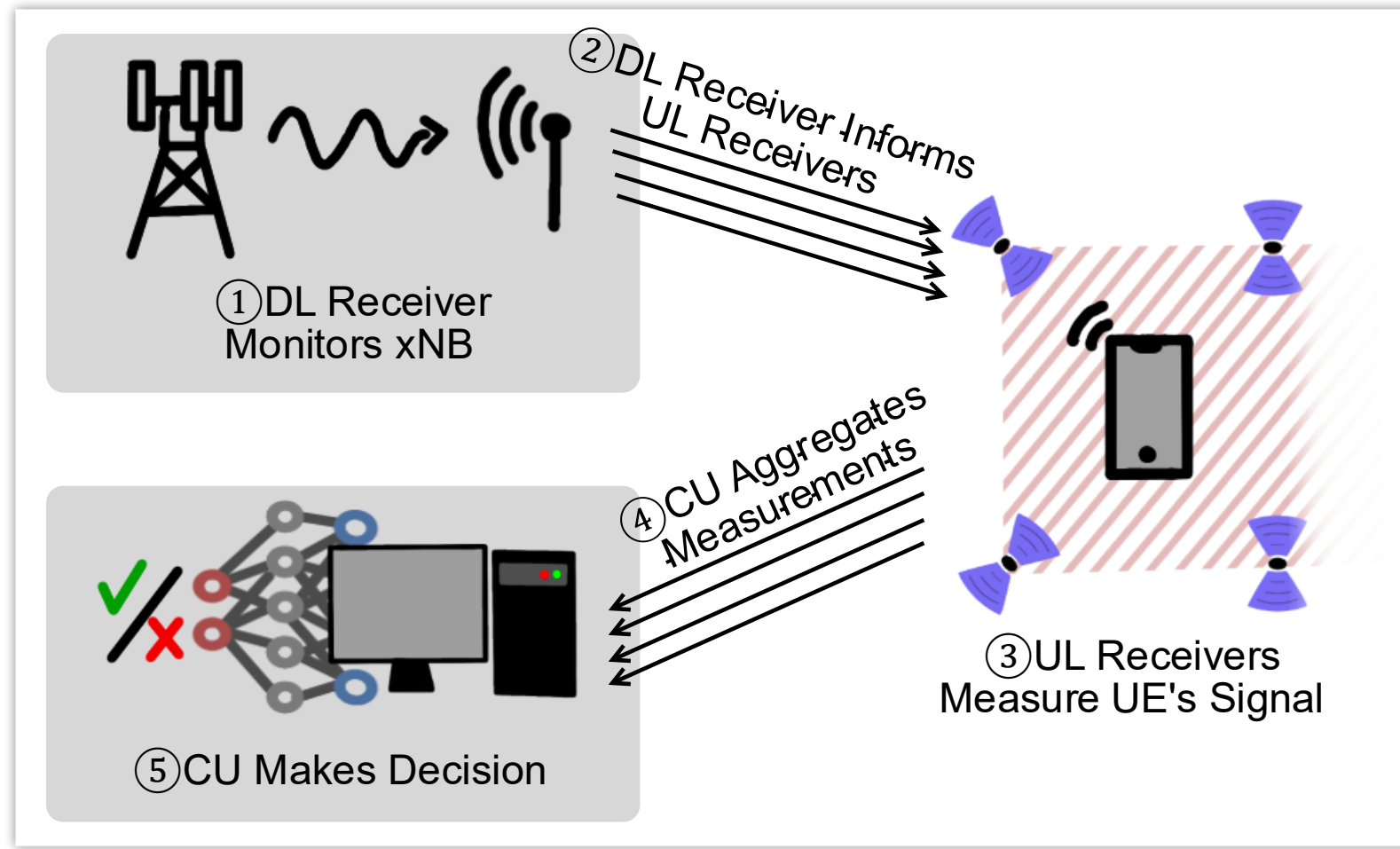
Connection Establishment



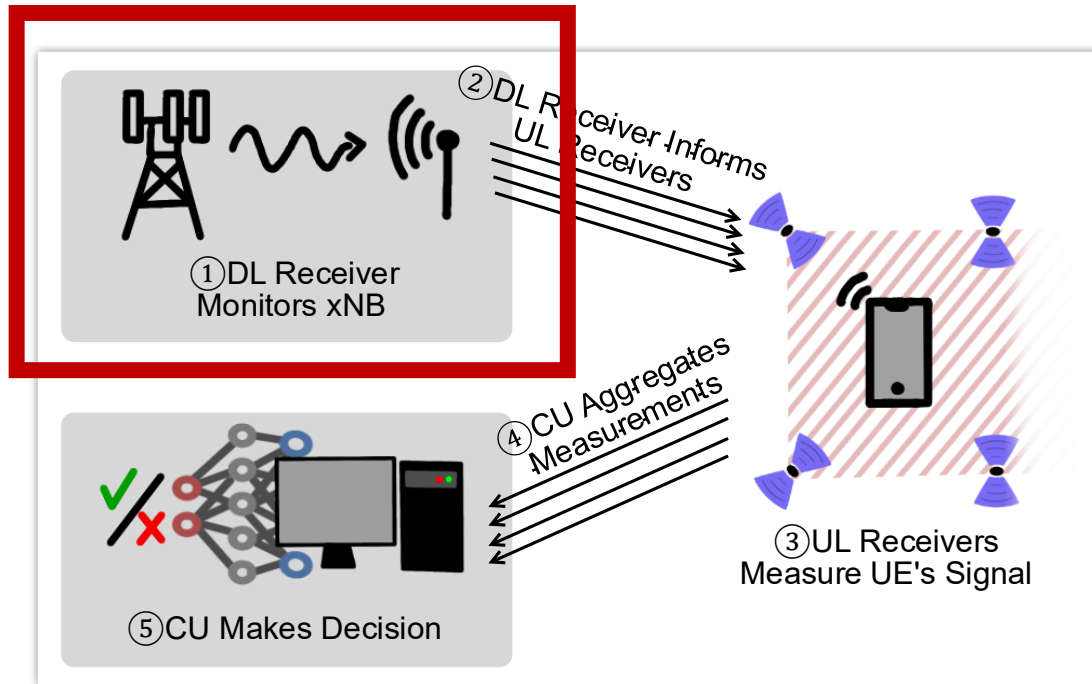
Connection established in this step!

This message has to be neutralized

System Design

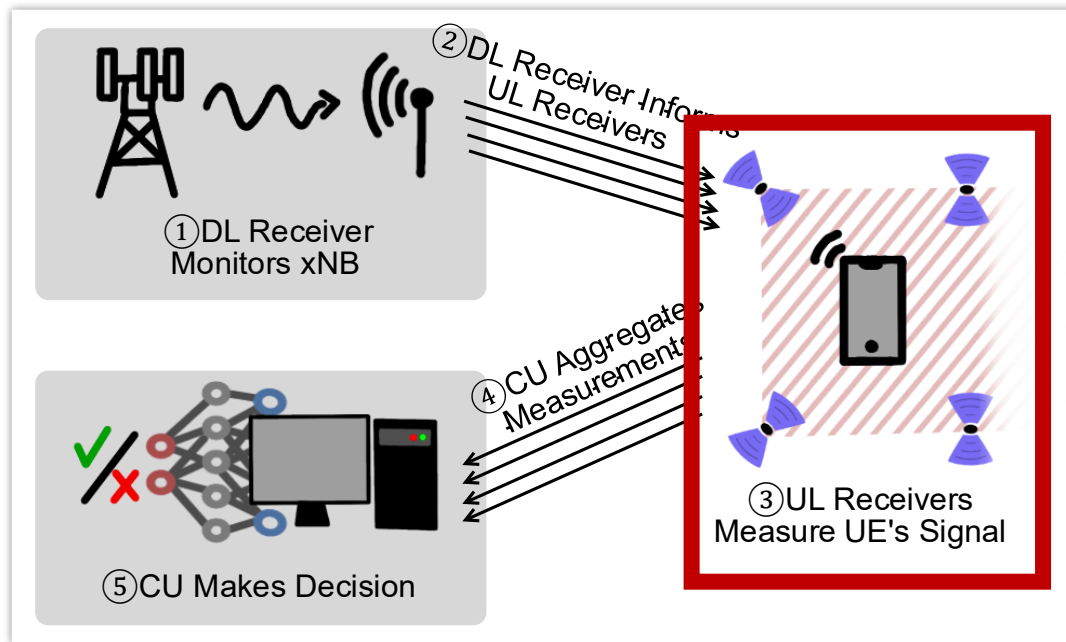


Downlink Reception



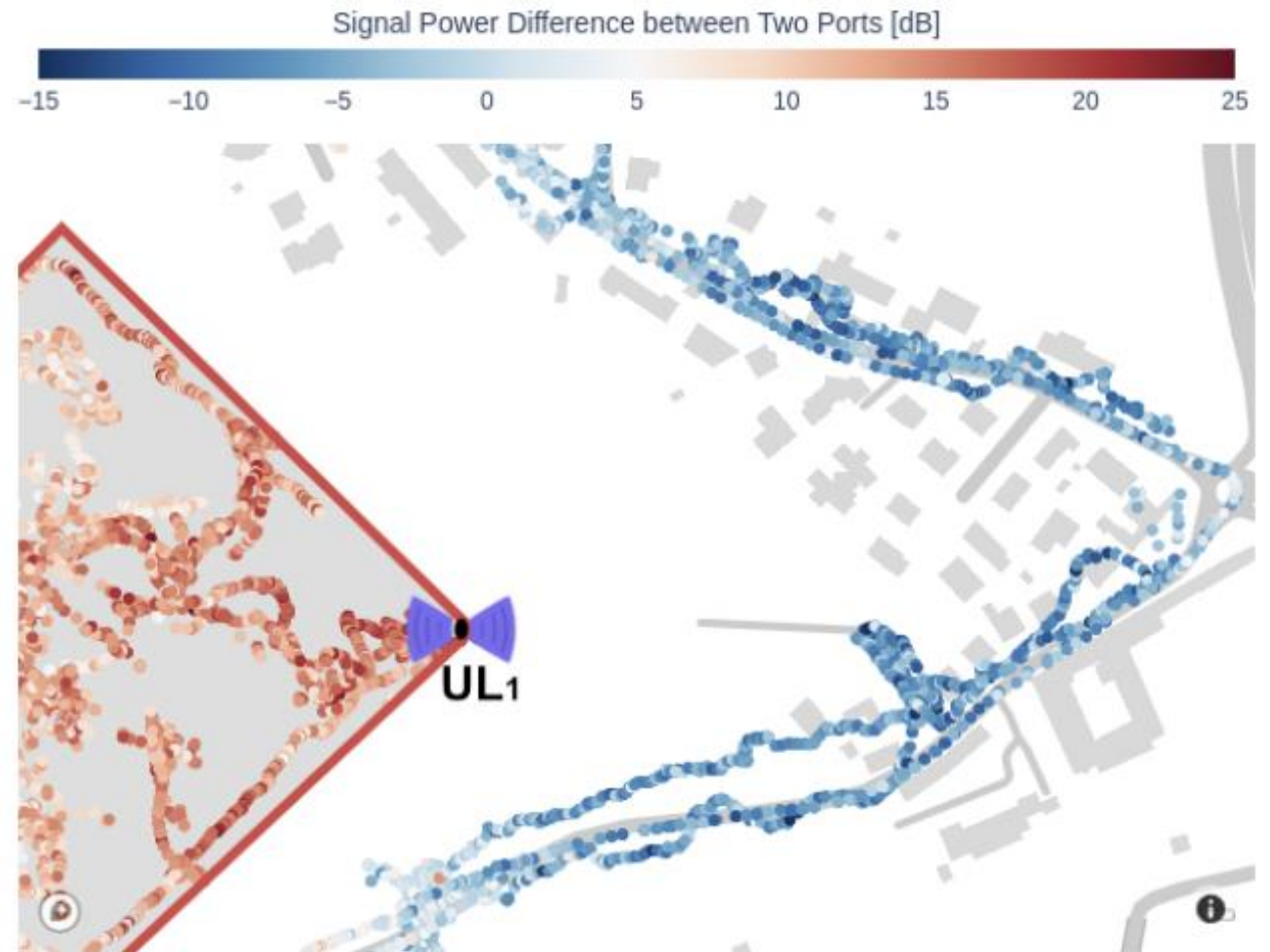
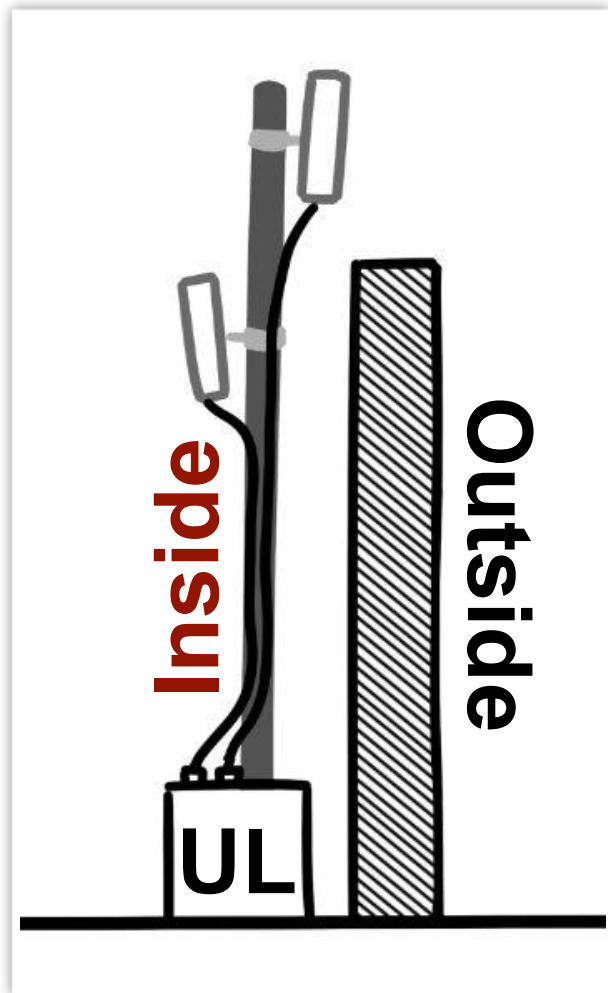
- Synchronized with all base station in the vicinity
- Detection of new connection based on RAR
- Distribution of uplink allocations to UL receivers

Uplink Measurement

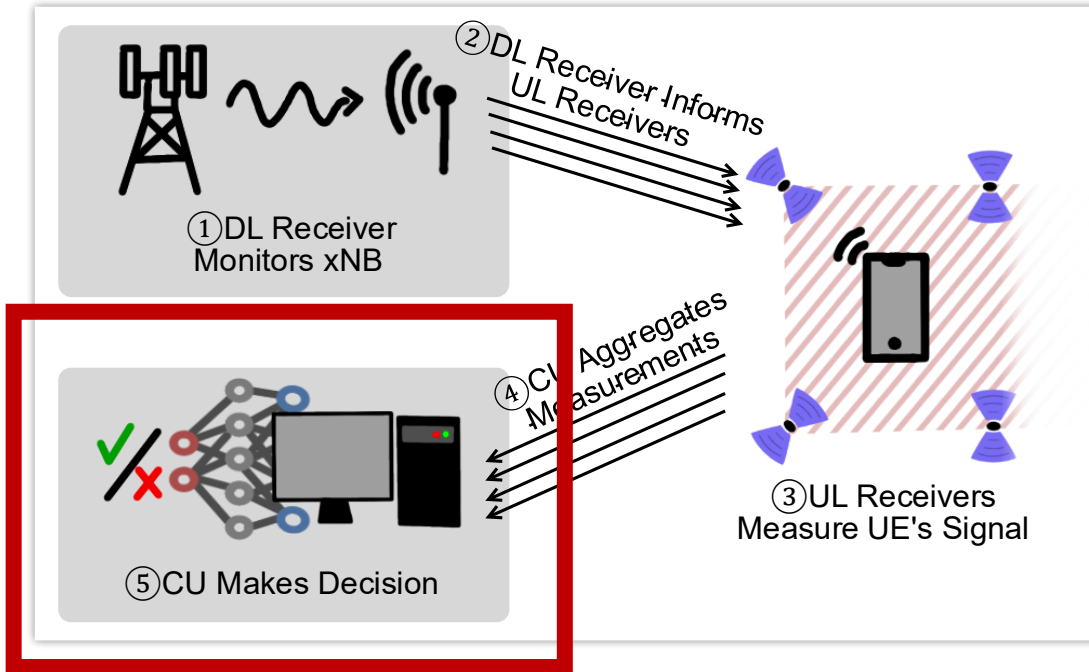


- Monitoring all uplink bands at the same time
- Synchronized receivers with WhiteRabbit
- Signal power & signal quality features based on correlation of reference symbols

Opposite Facing Antennas

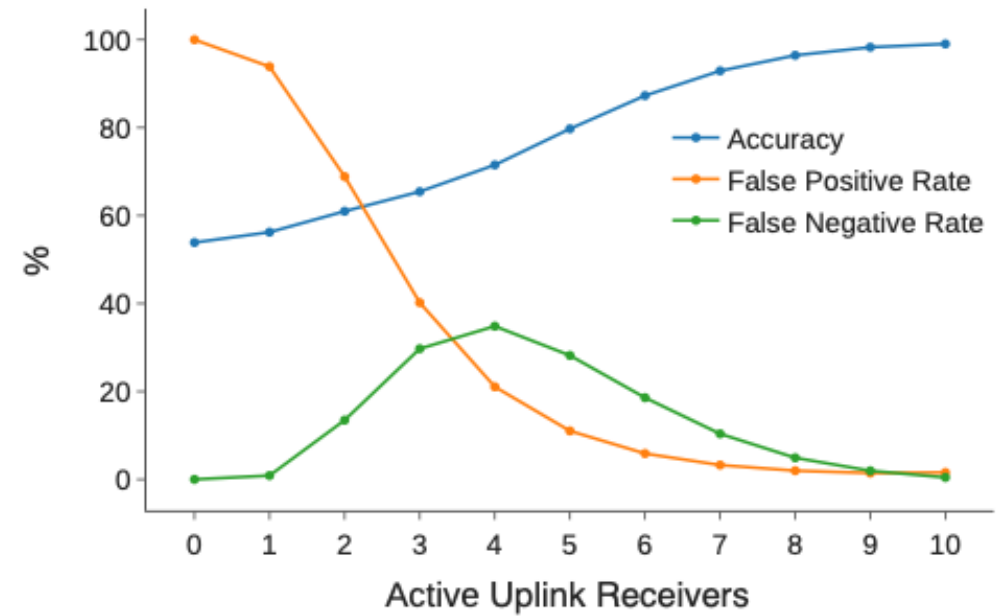
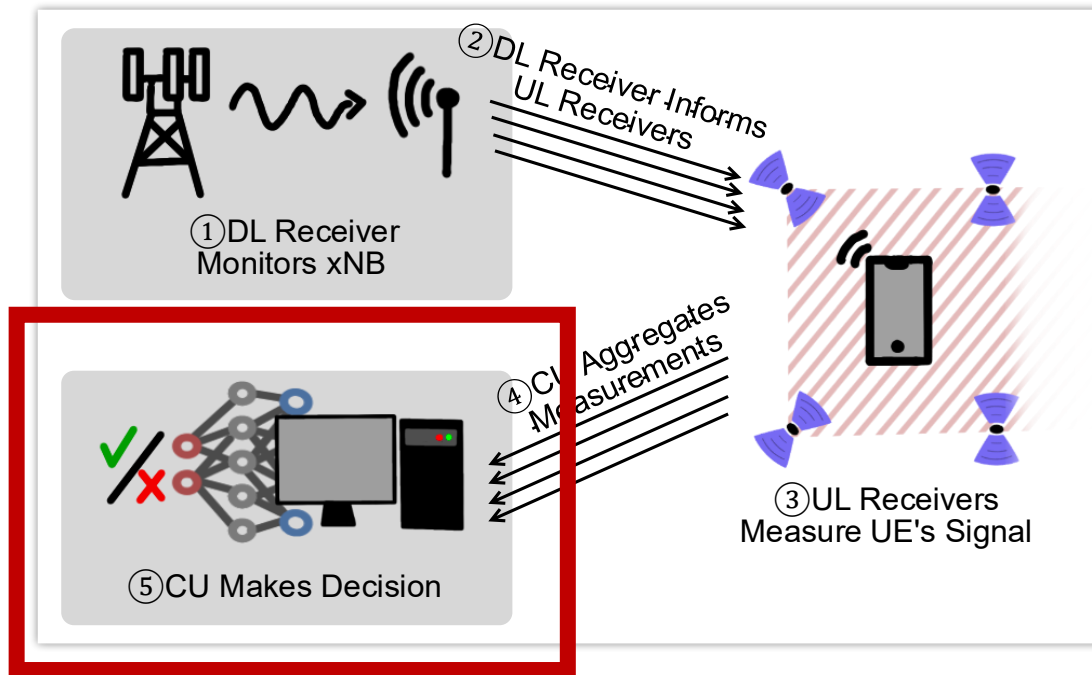


Inference

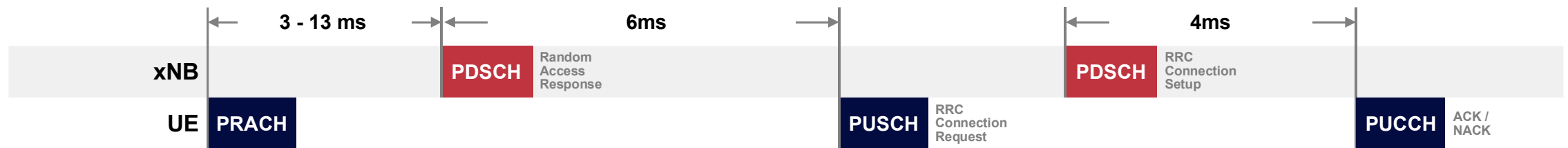


- Collect all measurement over ZMQ
- Robust ML model which combines results from UL Receivers
- Large dataset consisting of >170k connections

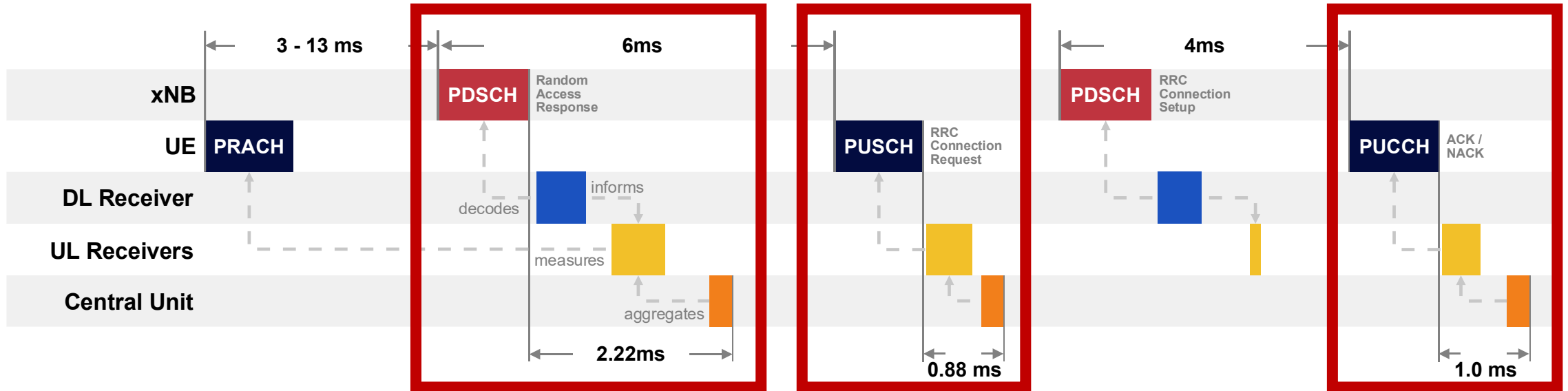
Inference



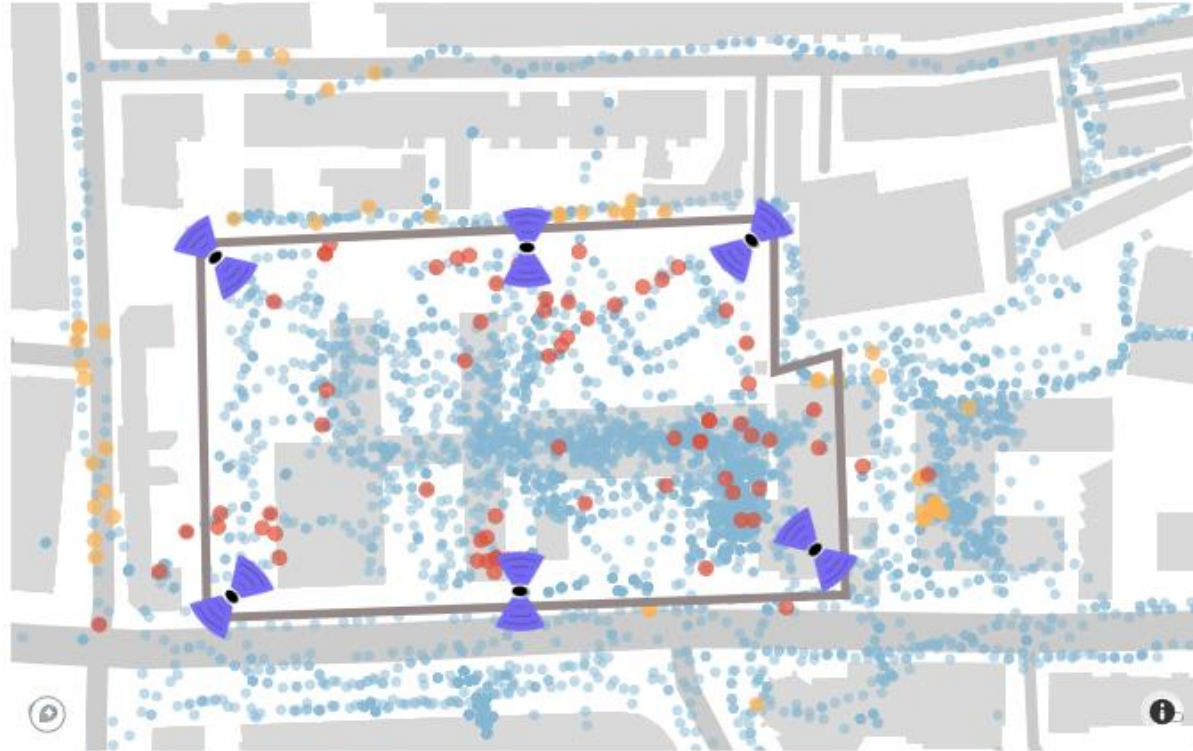
Latency Measurements in Real-World Deployment



Latency Measurements in Real-World Deployment



Evaluation in Real-World Conditions: First Deployment



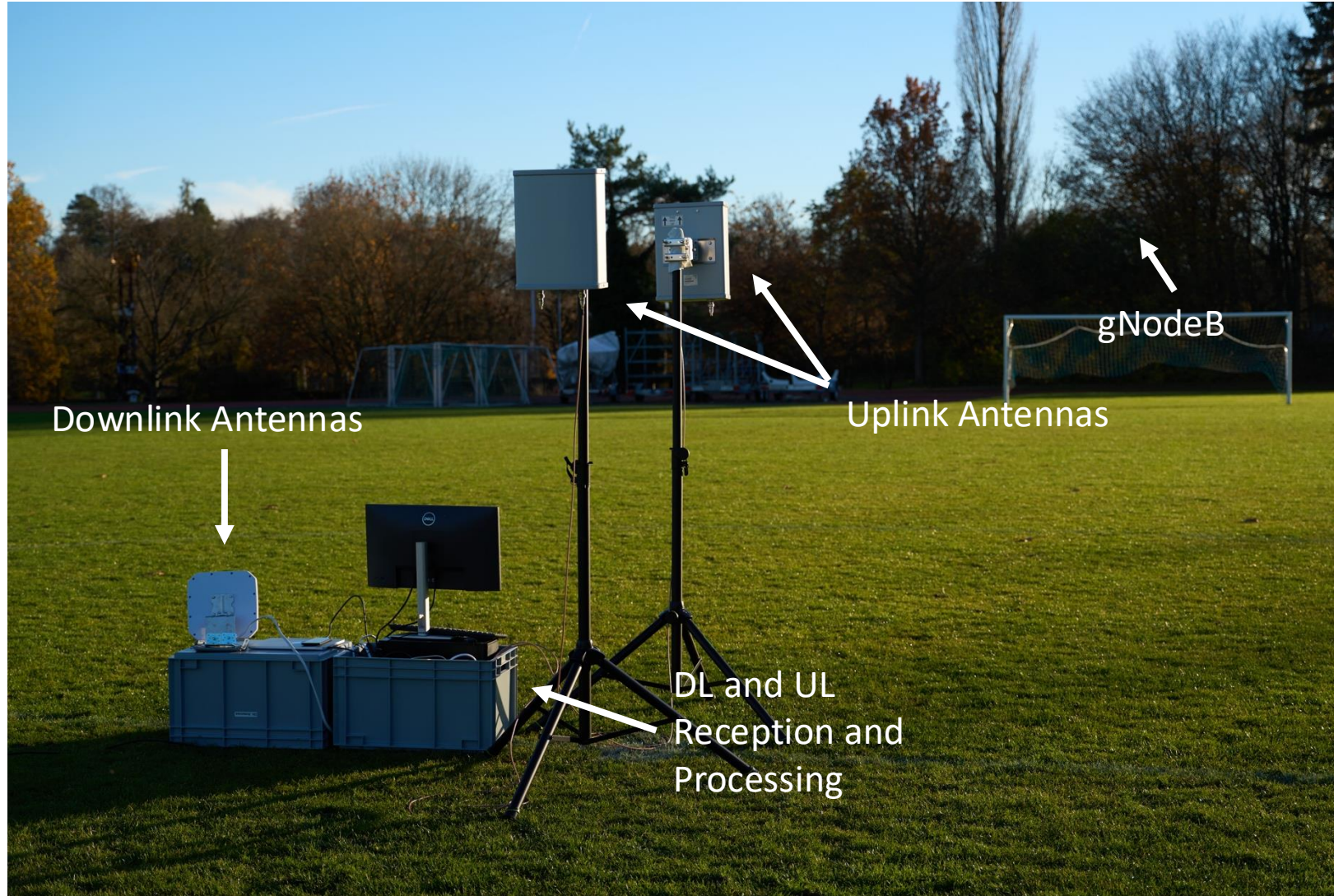
- Urban area, $\sim 100 \times 200$ meters, single multi-story building
- 1 downlink receiver + 6 uplink receivers
- COTS SDRs
- Inside/outside separation as little as 3 meters
- 97.42% geofencing accuracy

Evaluation in Real-World Conditions: Second Deployment

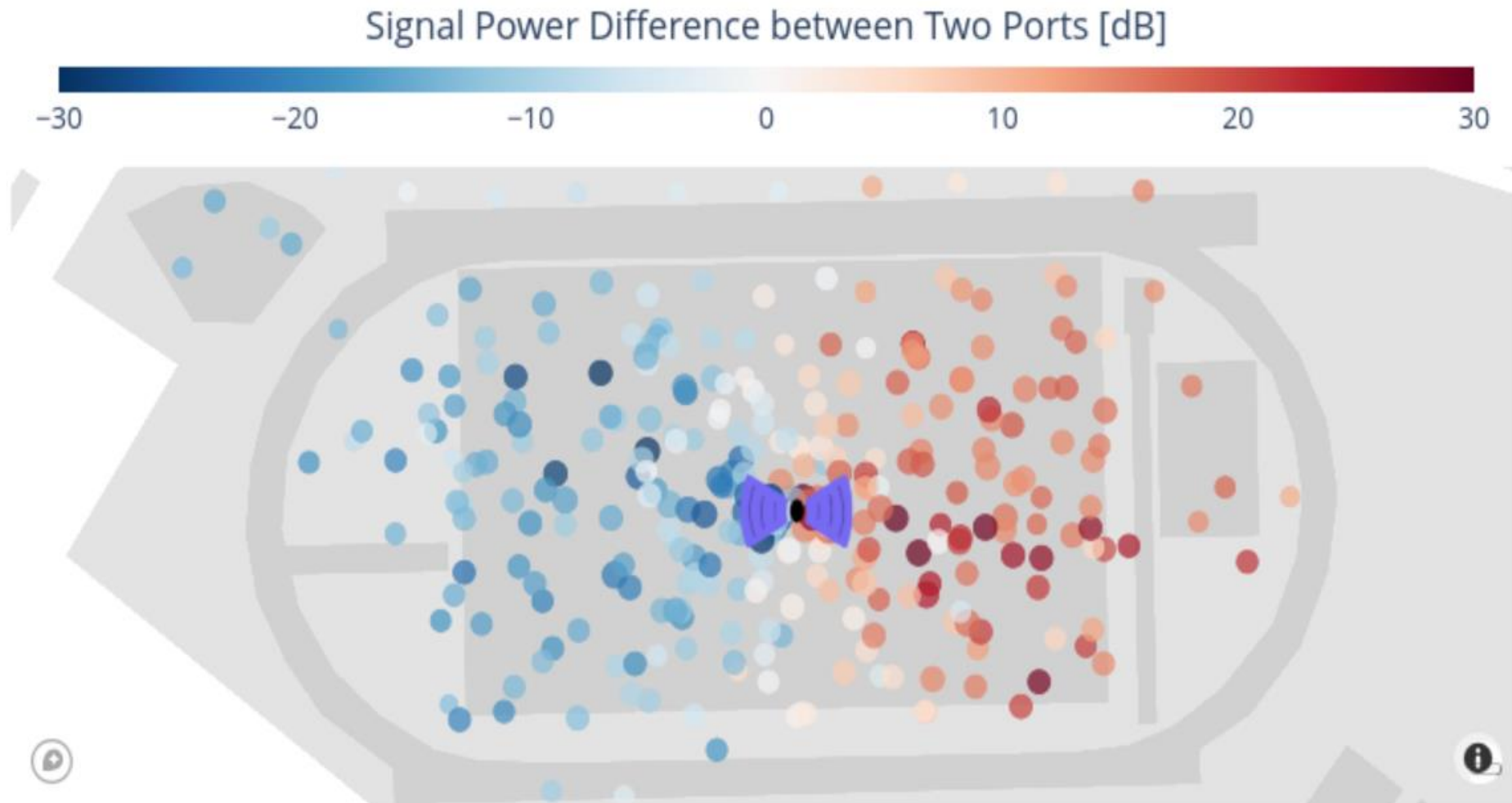
		Actual		
		Inside	Outside	
Predicted	Inside	10442	30	Acc = 99.66%
	Outside	43	10796	FPR = 0.28%
				FNR = 0.41%

- Larger urban outskirts site, $\sim 250 \times 250$ meters, multiple buildings
- 4 downlink receiver + 10 uplink receivers
- Custom-built SDRs
- Monitored 100+ cells across 4 operators simultaneously

Tests in Real-World 5G SA Deployment



Tests in 5G Deployment



Thank you! Questions?



Finding Phones Fast: Low-Latency and Scalable Monitoring of Cellular Communications in Sensitive Areas